

Coursework Assignment 1st sit

6E7Z2148 / Engineering Sustainability

This is an individual assignment for students on Level-7 of the following courses:-

MEng (Hons) Mechanical Engineering
MEng (Hons) Electrical and Electronic Engineering
MSc Digital Design and Manufacturing
MSc Engineering Project Management

Is your Pet Eco-friendly?

Nowadays, many pet owners become more conscious about the environment and impact their four-legged companions might have on the planet. However, choosing sustainable products can be sometimes tricky, as the market is saturated by pet toys made from materials that have a negative impact on the environment, such as nylon, polyester or conventional cotton. Choosing pet toys made from sustainable materials can help reduce plastic pollution and minimise its impact on the landfills and oceans, avoiding materials that can be potentially harmful or toxic not only for the environment but also your pets. Through this coursework, you will be able to contribute to the ambitious action plan to help decarbonise the pet-industry sector by investigating materials and methods of manufacturing sustainable pet toys through investigating materials design and manufacturing methods and the life cycle of the products.

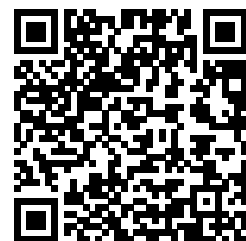
We hope that you enjoy doing this assignment. It is an opportunity for you to demonstrate how well you can apply your engineering knowledge to solve real world societal issues and help create a lower carbon future.



Links

The [Moodle Support Page](#) includes the support video, submission details, answers to frequently asked questions (FAQ), and the latest version of this document.

Software tools: This unit is heavily based on the use of Granta Edupack software. This [video-resource](#) allows you to access and use the software package required for the assessment in this unit when working off-campus.



Contacts

Lead Assessor: [Dr Justyna Kulczyk-Malecka](#)

Technical Assistance : Department Technical Services Engineering.Sci.TTL.Ass@mmu.ac.uk

Academic Assistance: Engineering Programme Support Tutors EngPST@mmu.ac.uk

Coursework Assignment 1st sit

6E7Z2148 / Engineering Sustainability

Contents

Workflow.....	3
Use of Artificial Intelligence (AI)	3
Task 1: Research the vast pool of products related to pet industry.....	4
Task 2: Analyse sustainable options	4
Task 3: Assess your proposal	4
Task 4: Write a report	5
Pass Criteria	5
What you need to do to achieve a pass at threshold level (Mark of 50%).....	5
What you need to do to achieve a pass above threshold level (Mark above 50%).....	6
Supporting Learning Resources and Activities.....	6
Assessment Notes.....	6
Deliverables.....	6
Marking.....	7
Formative Feedback.....	7
What is being assessed?	7
Marking Criteria	9
Re-assessment	10
Appendix	10
Summary of AHEP-4 Learning Outcomes Evidenced	10

Coursework Assignment 1st sit

6E7Z2148 / Engineering Sustainability

Workflow

Background Preparation	• Week 2: Carbon Literacy training - understanding carbon footprint. • Week 3: Carbon Literacy training - becoming carbon literate.
Task 1: Research pet-toys products.	• Week 1: Introduction to Sustainable Engineering. • Week 5: Guest Lecture: Sustainability in the car making industry.
Task 2: Analyse sustainability options	• Week 4: Granta Edupack materials selection. • Week 6: Granta Edupack process selection. • Week 7: Granta Edupack Eco Audit.
Task 3: Assess your proposal	• Week 8: Guest Lecture - Circular economy of sustainable packaging. • Week 9: Hydrogen - leading sustainable future. • Week 10: Renewable energy.
Documentation	• Week 11: Revision and final remarks.

Plan a realistic schedule of work to complete this assignment on time and to a high standard.

Recommended deadlines to complete each sub-task:

1. 20/10/2025 – research on sustainability, policies, risks and ethics.
2. 17/11/2025 – alternative materials and process selection.
3. 15/12/2025 – complete life cycle analysis.
4. 10/01/2026 – report writing.

The submission deadline for this assessment is **21:00** on **16/01/2026**. This assignment is for 100% of your module mark, and we expect you to spend up to 45 hours working on it.

The [Assessment Support video](#) will talk you through the workflow and help guide your plan for success.

Use of Artificial Intelligence (AI)

The use of generative AI is permitted in this assessment, so long as it is used in accordance with the instructions provided in the '[Are you allowed to use AI in assessments?](#)' section of the AI Literacy Rise Study Pack. All submitted work must be your own original content.

Coursework Assignment 1st sit

6E7Z2148 / Engineering Sustainability

Task 1: Research the vast pool of products related to pet industry

Initially you are required to undertake a thorough investigation of what materials are typically used for pet toys, focusing on multi-material/multi-component toy as your working example, such as teaser rods, electronic and interactive toys or toss and chase toys. What manufacturing routes are utilised to make these products? Are the products eco-friendly? Are they recycled, reclaimed, sustainable? Then gather information on the product's historic evolution in materials, specific functional requirements and research the market and life cycle context to the introduction of your report.

You must focus on detailed product specifications for this assessment, meaning you need to conduct your analyses on either two specific structural components of the item, or two separate products made out of different materials.

Task 2: Analyse sustainable options

In order to support the transition towards sustainable development and reduce waste production in our society, you are required to produce a technical report that:

- 1) Analyses and compares three different **materials** suitable for non-toxic pet-friendly products using taught software called Granta EduPack. To satisfy non-toxicity and sustainability agenda you should focus on three different material types:
 - recycled plastics
 - bio-degradable materials
 - responsibly sourced natural materials
- 2) Assesses the manufacturing of the product via a suitable **renewable energy source** (rather than conventional energy).

To achieve the above two points:

- Research and use Granta EduPack software to identify the properties of **materials of interest**, including recycling availability, process embodied energy and carbon footprint.
- Review the data of available **renewable energy** opportunities (showing **statistical** evidence for your choice, e.g. via published governmental reports by DEFRA or BEIS).
- Identify and implement relevant environmental policy guidance and build (via Granta EduPack) an **eco-audit** of your proposal.



You must make a final recommendation in your report on which scenario is the most eco-friendly and sustainable!

Task 3: Assess your proposal

Evaluate **economical**, **ethical** and **risk aspects** of your proposal.

Include critical **ethical** considerations to justify your proposal. You may refer to **environmental policy guidance** and **ethics for sustainable development**. In relation to the alternative materials and chosen strategy, you must conduct a Health and Safety appraisal for your proposed solution.

Coursework Assignment 1st sit

6E7Z2148 / Engineering Sustainability

Estimate the **economic feasibility** and **commercial risk** of your proposal, highlighting benefits and costs in your discussion. For your calculations, if relevant, use **units** as per examples below:

- Power (P) Generated in **kW**
- Investment Ratio (IR) in **£/kW**
- Operating Hours (OH) in **Hours/year**
- Useful Life (UL) in **Years**
- Operation and Maintenance Costs (O&M) in **£ * 10⁻³ /kWh**
- Tons of CO₂ avoided in total **CO₂/y**

Task 4: Write a report

Report format: You are required to present your work in the report format of **approx. 12 A4 pages** (font Arial, size 11 interline spacing 1.0, all margins at 2cm), **excluding references and appendices**.

The report must address the requirements/learning outcomes as per unit handbook available on Moodle.

Please use the [Report Template](#) available to download from Moodle. The information that you have gathered through your work on background research, investigations and recommendations may be gathered from the following sources:

- Secondary research (scientific magazines, publications, case studies, etc.)
- Granta Edupack
- Statistical analysis
- A review of the legal, ethical, sustainability, environmental and commercial constraints
- Government documents (strategies, targets, policies)

Using the [Cite Them Right Harvard Referencing](#), list all specific sources of information you have referred to in your work.

The research data for this should come from at least five formal scientific communications (books, conference/journal papers, patents, standards). A reference list consisting solely of websites/informal online communications is not acceptable. Websites are acceptable in certain circumstances, such as the websites of governing bodies (e.g., FIRA, UK Fire Regulations...), but it is preferable to cite specific documents located on these websites.

Appendix

This can be limitless, but should only contain relevant auxiliary tables, graphs, statistical data etc. You must explicitly refer to specific items in your appendix in the text body of your technical report, hence content should be adequately numbered. Data placed in the appendix that was not referred to in the main text is not considered as relevant.

Pass Criteria

What you need to do to achieve a pass at threshold level (Mark of 50%)

To pass this unit you need to show evidence that you have satisfactorily completed what is described in Task 1 and 2. This means that you have successfully selected (using Granta Edupack software) a suitable alternative material for your target product in relation to its specific application and designed the renewable energy plant that will satisfy the energy demand of the manufacturing plant of that item.

Coursework Assignment 1st sit

6E7Z2148 / Engineering Sustainability

What you need to do to achieve a pass above threshold level (Mark above 50%)

To achieve above 50%, you must ensure your proposal covering all points described in Task 3. These points are explicitly stated and exemplified by four sub-headings in Section 4 of the coursework report template. They are specifically: 4.5.1. Economic feasibility & commercial risk, 4.5.2. Environmental considerations, 4.5.3. H&S appraisal, 4.5.4. Ethical considerations.

Supporting Learning Resources and Activities

- [Proposal evaluation example](#)
- [Good Practice - example of a report](#)
- [Coursework preparation](#)

Assessment Notes

Your assessments can be used as evidence of your competencies to a potential employer. To convey your professionalism, formal reports should at the minimum adhere to the following:

- The report should be typed using a Word software (make use of the review tools to check spelling and grammar).
- Include a title sheet with your name and MMU ID clearly visible.
- Ensure all digital figures and images are clear and legible (not pixelated, correct aspect ratio).
- Ensure the source of all figures and images that you have not personally created are referenced.
- Ensure that all the figures and tables contain captions describing the figure/table.
- Ensure all equations at a minimum are typed or use an equation editor (i.e. do not draw equation by hand, then scan and paste them in the text).

Deliverables

The main deliverables are **Coursework report**, **Grata Edupack files** for material/process selection and **Eco Audit** analysis.

Your report should address the following key points:

- Abstract (max 200-250 words), should answer the following questions:
 - i. What is the purpose of the work?
 - ii. What activities you have done for this project?
 - iii. What were the main findings and conclusions reached as a result of this project?
- Introduction
 - i. Describe your starting point with the rationale for undertaking the work and the background to the subject.
 - ii. Clearly state the aim and objectives of the assignment task.
 - iii. Structure of the report-briefly explain how the rest of the report is organised.
- Literature review
 - i. This is a survey of publications (books, journals, authoritative websites, sometimes conference papers) reporting work that has already been done on the topic of your report.
 - ii. It should only include studies that have direct relevance to your research.
 - iii. Group your texts in themes. Write about each theme as a separate section, giving a critical summary of each piece of work and showing its relevance to your research.
 - iv. Conclude with how the review has informed your research.
- Methodology - describe the analysis techniques applied such as Granta Edupack software.
- Results and discussion
 - i. Break down the section into several sub-sections to help the reader to understand the work.

Coursework Assignment 1st sit

6E7Z2148 / Engineering Sustainability

- ii. Present the findings of your project/assignment task using text, graphs, tables or diagrams.
- iii. The quality of this section will have a large influence on your marks.
- iv. Writing in this section should be clear, factual and informative.
- v. This section must present the outcome of critical analysis of your results.
- Conclusions
 - i. Conclusions should be a short section with no new arguments or evidence.
 - ii. Sum up the main findings of your work.
 - iii. This section may also include suggestions for further work.
- References (excluded from page count) - the University Library has an online tutorial to help you find good quality reference sources and a guide on how to use them correctly in your document using the MMU Harvard referencing style.

Your report should include at least 15 references to support your design/conclusions, with at least two of each of the following types of reference (use the [Cite Them Right Harvard style](#)):

 - i. A published reference source in the University Library
 - ii. A web reference (all types of media)
 - iii. A technical reference source, e.g. manufacturer's datasheet, industrial standard, etc.
- Appendices (excluded from page count)
 - i. Appendices are not compulsory for the report but can help in better understanding it. Supporting (and relevant!) information that cannot be accommodated in the main report should be included in the Appendix.
 - ii. Create a sub-section for each appendix and refer to them where appropriate in the main text (e.g. *See Appendix A*).

Marking

Formative Feedback

Members of the teaching team will answer queries about the assignment during the timetabled Workshop classes and in the drop-in Surgery sessions (see your personal timetable for details). Advice given during these sessions and feedback on any work you present will not count towards your final grade.

Outside timetable sessions, please email questions to the Module Teaching Team, the Engineering Programme Support Tutors EngPST@mmu.ac.uk or your Personal Tutor. The teaching team will **not** use their office hours to provide one-to-one or small group assignment support, but a selection of emailed questions, and questions asked during the timetabled classes will be added to the assignment's [FAQ's](#) page.

What is being assessed?

Learning outcomes		Evidence
Module Learning Outcomes	1. Analyse methods to solve complex problems in renewable energy and to assess their limitations.	Report: statistical analysis, eco-audits, Granta Edupack on materials selection, etc.
	2. Defend a professional and ethical approach to sustainable engineering solutions.	Report: professional and ethical values for sustainable development
	3. Appraise relevant regulatory requirements	Report: environmental policies

Coursework Assignment 1st sit

6E7Z2148 / Engineering Sustainability

	governing sustainable engineering activities.	
	4. Evaluate risk issues in the context of sustainable engineering, including health & safety, environmental and commercial risk.	Report: economic feasibility, tonnage CO2, H&S limitations, etc.
Assessment Criteria	USD1 Apply a comprehensive knowledge of mathematics, statistics, natural science and engineering principles to the solution of complex problems. Much of the knowledge will be at the forefront of the particular subject of study and informed by a critical awareness of new developments and the wider context of engineering. Select and apply appropriate materials, equipment, engineering technologies and processes, recognising their limitations. <i>AHEP4 outcomes evidenced: M1, M13</i>	
	USD3 Communicate effectively on complex engineering matters with technical and non-technical audiences, evaluating the effectiveness of the methods used. <i>AHEP4 outcomes evidenced: M17</i>	
	USD7 Design solutions for complex problems that evidence some originality and meet a combination of societal, user, business and customer needs as appropriate. This will involve consideration of applicable health & safety, diversity, inclusion, cultural, societal, environmental and commercial matters, codes of practice and industry standards. Evaluate the environmental and societal impact of solutions to complex problems (to include the entire life-cycle of a product or process) and minimise adverse impacts. <i>AHEP4 outcomes evidenced: M5, M7</i>	

To pass this assessment you have to fully achieve the module learning outcomes by completing all the tasks and submitting the deliverables to an adequate standard. Since your degree is accredited, you also need to demonstrate evidence of obtaining [UK SPEC AHEP-4](#) outcomes at pass threshold levels. These are listed in full within the Appendix of this assessment brief.

Your grade will be determined by how well you meet the assessment criteria (see the detailed grading criteria for this assignment, below).

This assignment will be marked out of 100 and contributes 100% of overall module grade.

Coursework Assignment 1st sit

6E7Z2148 / Engineering Sustainability

Marking Criteria

Deliverable →	USD1 Report	USD3 Report	USD7 Report
Weighting →	20%	30%	50%
Grade range ↓	Apply a comprehensive knowledge of mathematics, statistics, natural science and engineering principles to the solution of complex problems. Much of the knowledge will be at the forefront of the particular subject of study and informed by a critical awareness of new developments and the wider context of engineering. Select and apply appropriate materials, equipment, engineering technologies and processes, recognising their limitations.	Communicate effectively on complex engineering matters with technical and non-technical audiences, evaluating the effectiveness of the methods used.	Design solutions for complex problems that evidence some originality and meet a combination of societal, user, business and customer needs as appropriate. This will involve consideration of applicable health & safety, diversity, inclusion, cultural, societal, environmental and commercial matters, codes of practice and industry standards. Evaluate the environmental and societal impact of solutions to complex problems (to include the entire life-cycle of a product or process) and minimise adverse impacts.
86%-100%	The novel and complex engineering principles are critically reviewed in relation to sustainable product development, materials and process selection and product life cycle analysis used in theory and practice. Original solutions are generated and clearly expressed.	A complex and innovative sustainable solution is designed, planned and carried out meticulously to gather and synthesise useful information from a wide range of appropriate secondary sources to produce original outcomes of publishable standards.	The Economic, Legal, Social, Ethical & Environmental aspects of the proposed solution are critically evaluated in developing action plans, articulating conclusions and making recommendations of relevance to theoretical development and/or practical application.
70%-85%	Most of the novel and complex engineering principles are reviewed in relation to sustainable product development, materials and process selection and product life cycle analysis used in theory and practice. Original solutions are generated and clearly expressed.	A complex sustainable solution is designed, planned and carried out thoroughly to gather useful information from a wide range of appropriate secondary sources and synthesise the results to produce workable outcomes.	The Economic, Legal, Social, Ethical & Environmental aspects of the proposed solution are considered in developing action plans, articulating conclusions and making recommendations of relevance to theoretical development and/or practical application.
60%-69%	An acceptable selection of the novel and complex engineering principles are reviewed in relation to sustainable product development, materials and process selection and product life cycle analysis used in theory and practice. Original solutions are generated and expressed.	A sustainable solution is carefully planned and carried out to gather useful information from appropriate secondary sources and synthesise the results.	The Economic, Legal, Social, Ethical & Environmental aspects of the proposed solution are analysed carefully in drawing conclusions and making recommendations.
50%-59%	A limited number of the novel and complex engineering principles are reviewed in relation to sustainable product development, materials and process selection in product design and product life cycle analysis used in theory and practice. Original solutions are generated.	A sustainable solution is planned and carried out to gather information from appropriate secondary sources and synthesise the results.	The Economic, Legal, Social, Ethical & Environmental aspects of the proposed solution are considered in drawing conclusions and making recommendations.
40%-49%	An extremely limited number of the novel and complex engineering principles are reviewed in relation to sustainable product development, materials and process selection in product design and product life cycle	A limited attempt to plan and/or carry out a sustainable solution which gather information from appropriate secondary sources.	There is limited or incorrect identification of the Economic, Legal, Social, Ethical & Environmental aspects of the proposed solution in drawing conclusions and making recommendations.

Coursework Assignment 1st sit

6E7Z2148 / Engineering Sustainability

	analysis used in theory and practice. Solutions are generated.		
35%-39%	An unacceptably limited number of the novel and complex engineering principles are reviewed in relation to sustainable product development, materials and process selection in product design and product life cycle analysis used in theory and practice.	An unacceptably limited attempt to plan and/or carry out a sustainable solution which gather information from appropriate secondary sources.	An unacceptably limited or incorrect identification of t Economic, Legal, Social, Ethical & Environmental aspects of the proposed solution in drawing conclusions and making recommendations.
20%-34%	The engineering principles are not reviewed in relation to sustainable product development, materials and process selection in product design and product life cycle analysis used in theory and practice.	Little attempt to plan and/or carry out a sustainable solution which gather information from appropriate secondary sources.	There is little or no identification of Economic, Legal, Social, Ethical & Environmental aspects of the proposed solution in drawing conclusions and making recommendations.
0%-19%	The engineering principles are not considered in sustainable product development or materials and process selection in product design and product life cycle analysis used in theory and practice.	No attempt to plan and/or carry out a sustainable solution which gather information from appropriate secondary sources.	There is no identification of Economic, Legal, Social, Ethical & Environmental aspects of the proposed solution in drawing conclusions and making recommendations.

Re-assessment

Note that opportunities to repeat assignments are not automatically provided, and when they are, the marks obtained may be capped. For more information about the regulations, please contact the [Student Hub](#).

If you think you are at risk of failing this assignment, please seek advice as soon as possible inform your Personal Tutor or the [Support Tutors](#) – with enough notice, there are many ways in which we can support you and help you to get back on track.

Appendix

Summary of AHEP-4 Learning Outcomes Evidenced

M1.	Apply a comprehensive knowledge of mathematics, statistics, natural science and engineering principles to the solution of complex problems. Much of the knowledge will be at the forefront of the particular subject of study and informed by a critical awareness of new developments and the wider context of engineering.
M5.	Design solutions for complex problems that evidence some originality and meet a combination of societal, user, business and customer needs as appropriate. This will involve consideration of applicable health and safety, diversity, inclusion, cultural, societal, environmental and commercial matters, codes of practice and industry standards.
M7.	Evaluate the environmental and societal impact of solutions to complex problems (to include the entire life-cycle of a product or process) and minimise adverse impacts.
M13.	Select and apply appropriate materials, equipment, engineering technologies and processes, recognising their limitations.
M17	Communicate effectively on complex engineering matters with technical and non- technical audiences, evaluating the effectiveness of the methods used.